Comparing groups

REFERENCE SHEET

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1. Math

Mean

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$\bar{y}$ is the sample mean, which is calculated as the sum of all observations y_i divided by the number of observations (n).

Median

The median is determined by finding the middle value. If n is odd, the median is the middle value in the dataset. If n is even, the median is the mean of the two middle values.

Standard deviation

$$s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$$

s is the sample standard deviation, which is calculated as the square root of the sum of the squared difference between an observation y_i and the sample mean $\bar{y}$ divided by the number of observations n minus 1.

Variance

$$s^2 = \frac{\sum(y_i - \bar{y})^2}{n-1}$$

Variance is calculated as the standard deviation squared. Note that the units of variance are the units squared (e.g. g^2, m^2, ppt^2).

Note that if you're describing spread in writing, it's usually best to go with standard deviation since the units are the same as the variable.

2. Calculating central tendency and spread

Prior to visualizing, it's sometimes helpful to actually calculate the metrics of central tendency (mean and median) and spread (standard deviation and variance) along with the sample size.

Site	Mean	Median	Standard deviation	Variance	Number of surveys
Phelps Bridge	3.7	2.2	5.3	27.9	234
East Channel	19.0	19.7	16.2	262.3	180
Venoco Bridge	22.7	23.4	18.3	336.0	242

3. Visualizing distributions

The purpose of visualizing distributions is to represent the distribution of variable of interest.

Note that the interpretation of these plots is more or less the same: Most salinity measurements from Phelps Bridge are low. Most salinity measurements from East Channel and Venoco Bridge are below 40 ppt.

Plots to start with

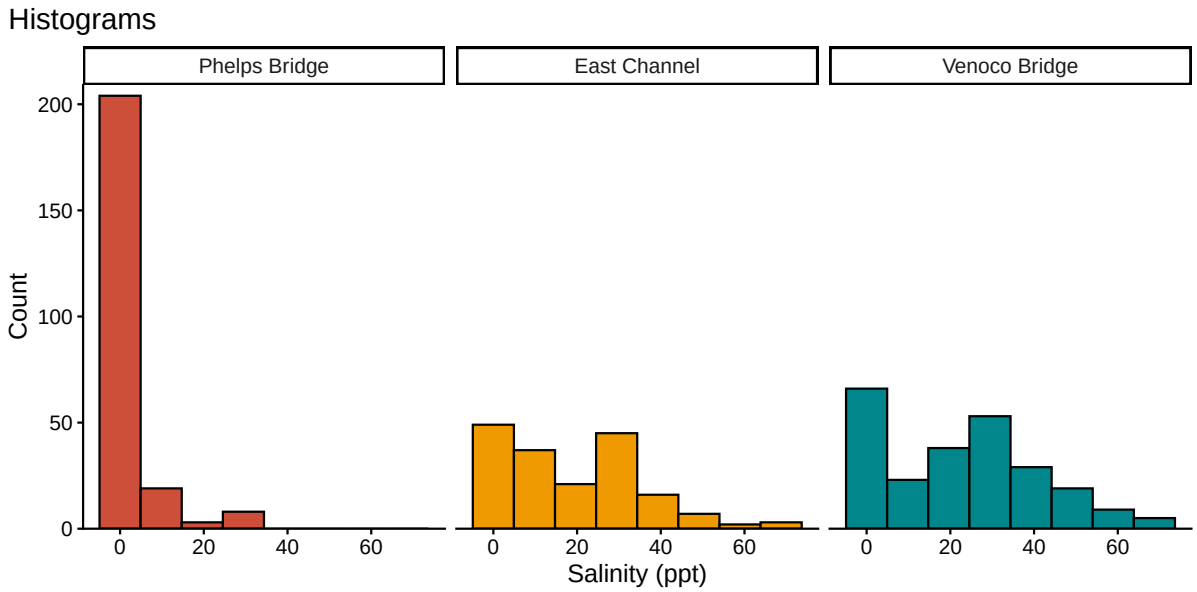
Histograms

Components

- x-axis: *continuous* variable of interest
- y-axis: counts
- Column height represents number of observations falling into a given bin range on the x-axis

Pay attention to:

- the shape of the distribution
- the center of the distribution
- the peak(s) of the distribution



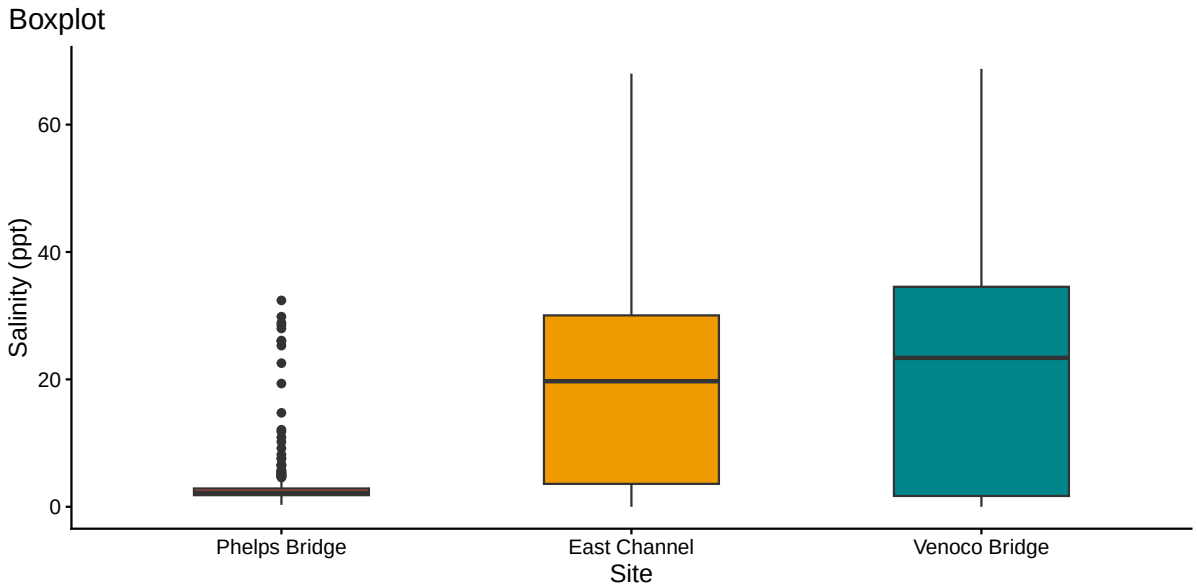
Potential caption:

Figure 1. Distributions of salinity differ across sites. Distributions of salinity (parts per thousand) shown in histograms for each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge). Bars represent counts of observations falling in each bin along the x-axis.

Boxplots

Components:

- x-axis: categories (levels in a factor)
- y-axis: variable of interest
- middle line: median
- bottom of box: first quartile (25th percentile)
- top of box: third quartile (75th percentile)
- whiskers: 1.5 $\times$ interquartile range
- points: observation values outside 1.5 $\times$ interquartile range



Potential captions:

Figure 2. Median salinity differs across sites. Boxplots representing median (middle line) salinity (parts per thousand) and 25th and 75th percentiles at each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge). Whiskers represent observations within 1.5 × interquartile range, and points represent observations outside the range.

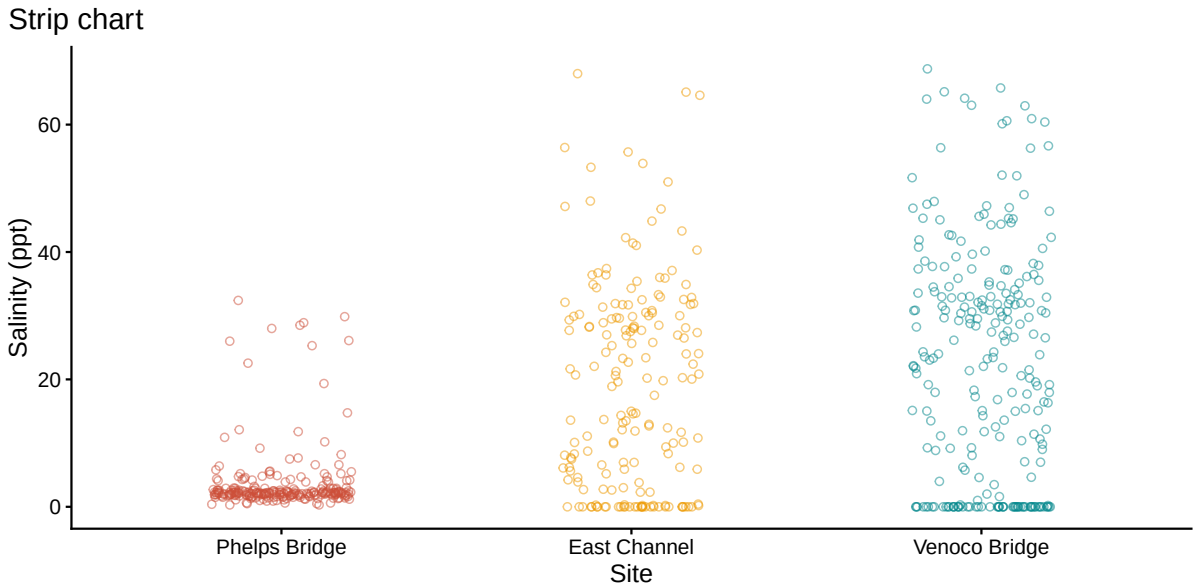
Strip charts

Components:

- x-axis: categories (levels in a factor)
- y-axis: variable of interest
- geometry: jittered points

Pay attention to:

- the clusters of points, if any
- position of points along the y-axis



Potential captions:

Figure 3. Median salinity differs across sites. Boxplots representing median (middle line) salinity (parts per thousand) and 25th and 75th percentiles at each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge). Whiskers represent observations within 1.5 × interquartile range, and points represent observations outside the range.

Taking things a step further

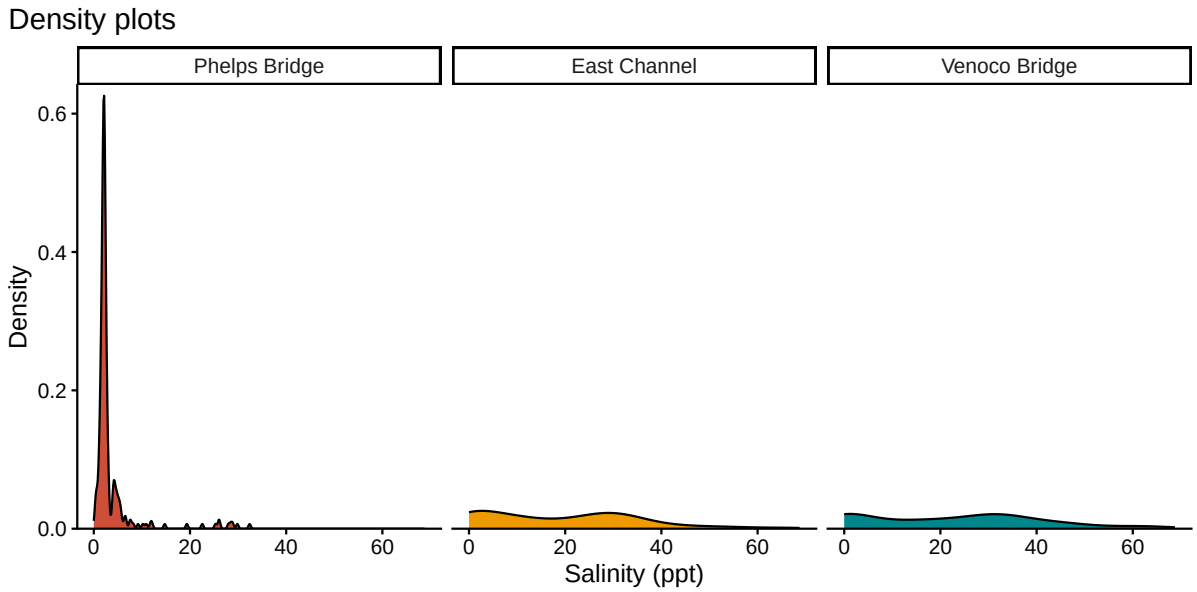
Density plots

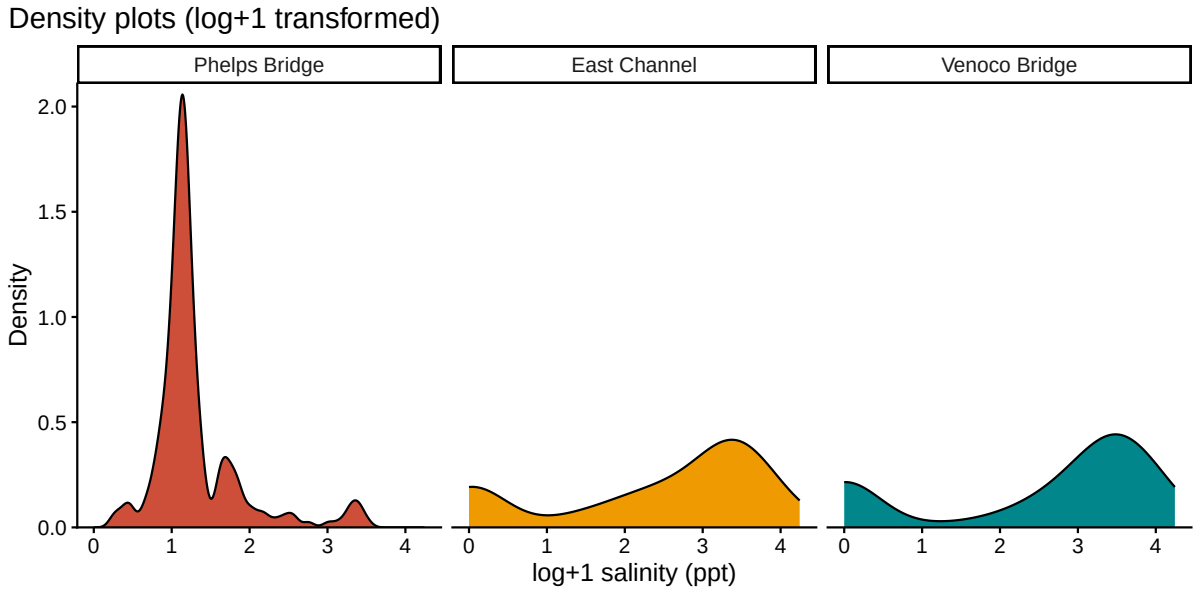
Components:

- x-axis: continuous variable of interest
- y-axis: probability density
- smoothed function representing the shape of the distribution
- note: area under that smoothed function = 1

Pay attention to:

- the shape of the **distribution**
- the center of the distribution
- the peak(s) of the distribution





Potential captions:

Figure 4. Distributions of salinity differ across sites. Distributions of salinity (parts per thousand) shown in density plots for each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge).

Figure 5. Distributions of salinity differ across sites. Distributions of log+1 transformed salinity (parts per thousand) shown in density plots for each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge).

> **Use log+1 transformation with caution!**
>
> The transformation can help with visualization (i.e. make it easier to pick out patterns), but can be tricky to interpret/analyze. Proceed with caution and ask yourself whether you really *need* the transformation to see patterns.

Violin charts

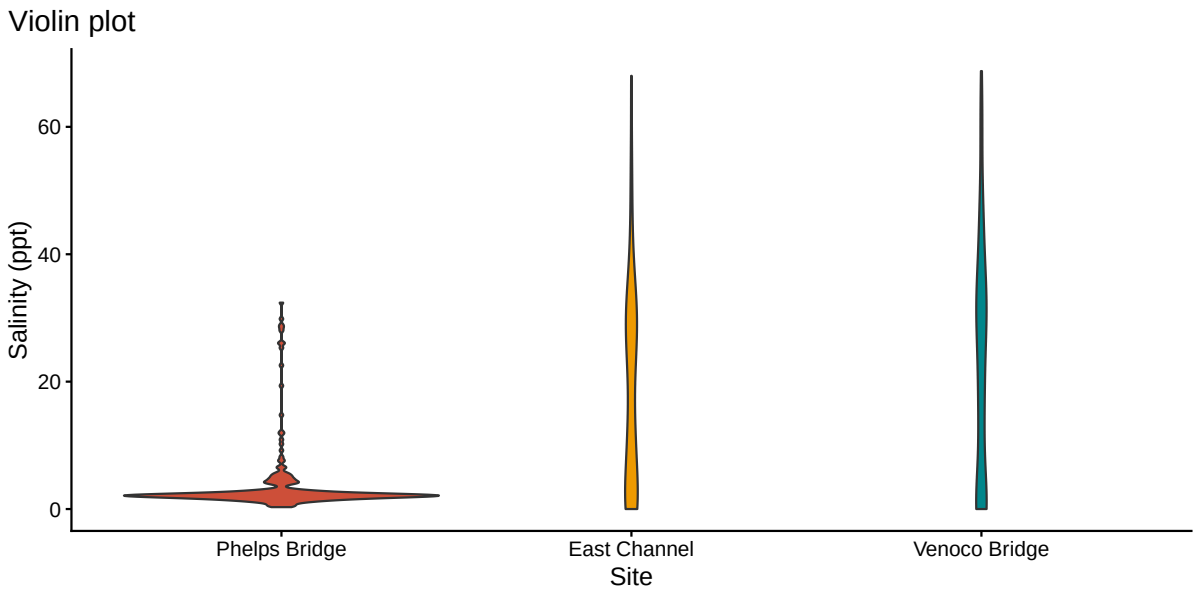
Note: violin charts are density plots that are mirrored.

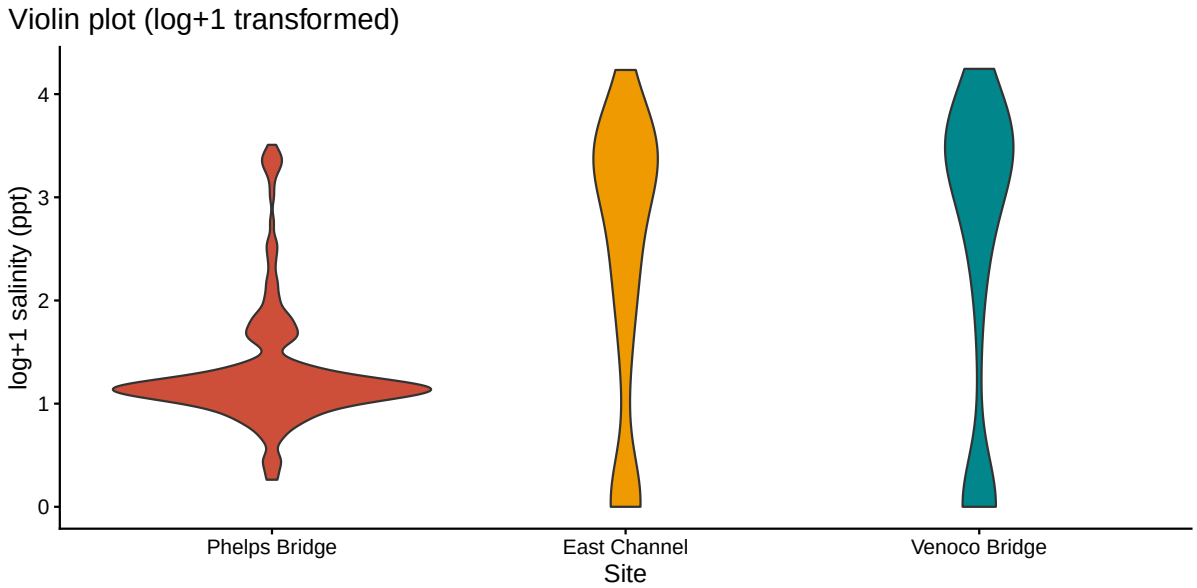
Components:

- x-axis: categories (levels in a factor)
- y-axis: variable of interest
- geometry: jittered points

Pay attention to:

- wide or narrow parts of the violin
- position of wide or narrow parts along y-axis





Potential captions:

Figure 6. Distributions of salinity differ across sites. Distributions of salinity (parts per thousand) shown in violin plots for each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge).

Figure 7. Distributions of salinity differ across sites. Distributions of log+1 transformed salinity (parts per thousand) shown in violin plots for each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge).

> **ℹ Plot pitfalls**
>
> Density plots, violin plots, and boxplots are appropriate if you have many observations per group (or level in a factor). If you have few observations, those plots will either
>
> a) hide the fact that you have few observations (as in boxplots),
> b) suggest the existence of values where there aren't any (as in density or violin plots)
> c) or some combination of both.
>
> There's no hard and fast rule for the number of observations that are appropriate. However, the solution to these problems is almost always to *show the underlying data* in a jitter or something similar.

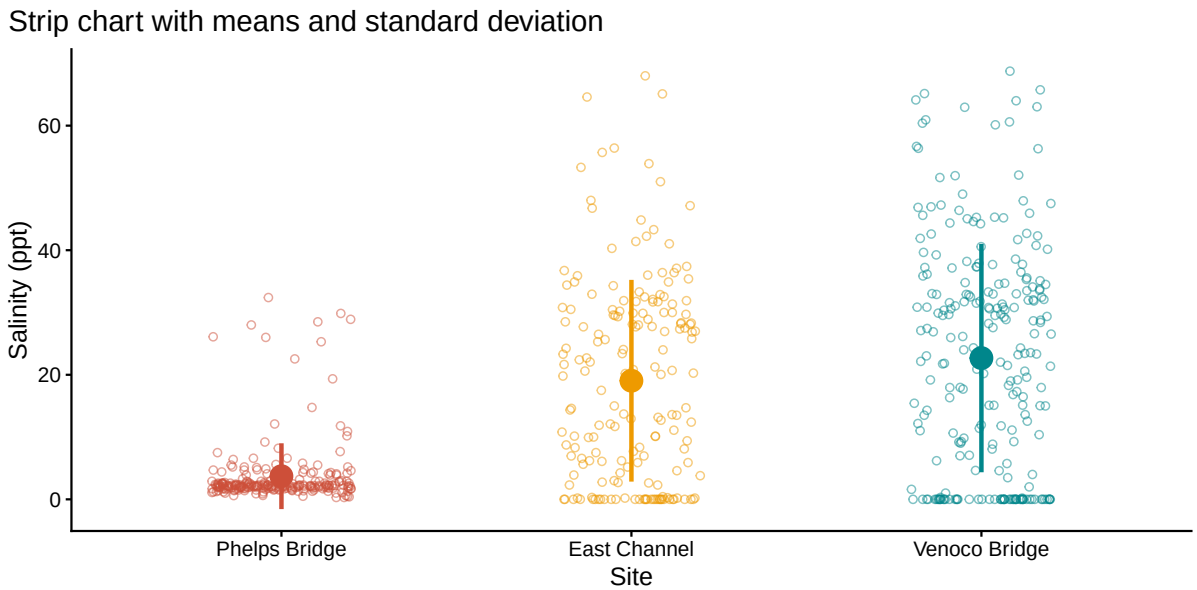
4. Visualizing central tendency and spread

Medians

Refer to the section on boxplots - these incorporate medians along with spread (quantiles).

Means

With standard deviation:



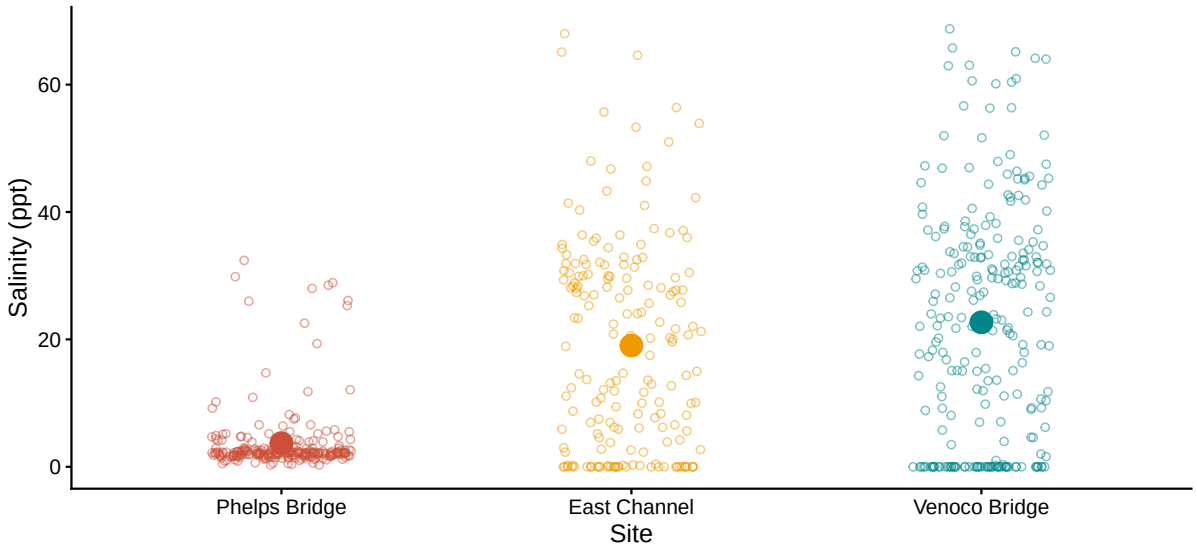
With standard error:

Note that standard error is calculated as:

$$\frac{s}{\sqrt{n}}$$

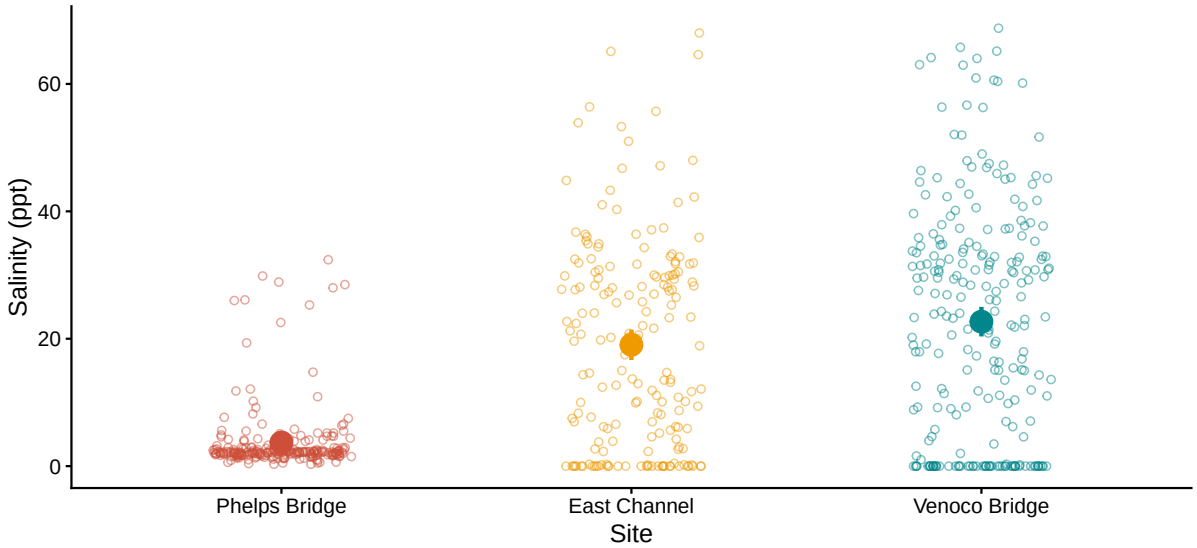
where s is the standard devation and n is the number of observations in the sample.

Strip chart with means and SE



With 95% confidence interval:

Strip chart with means and 95% confidence interval



> **Be explicit about what whiskers mean!**
>
> These plots all use the same geometry (`pointrange`) to represent the means and whiskers, but all three plots use whiskers to represent different values.
> Whenever making a plot with means and whiskers, it's *very* important to be explicit about what values those whiskers represent in the axis label and/or the caption.

Potential caption:

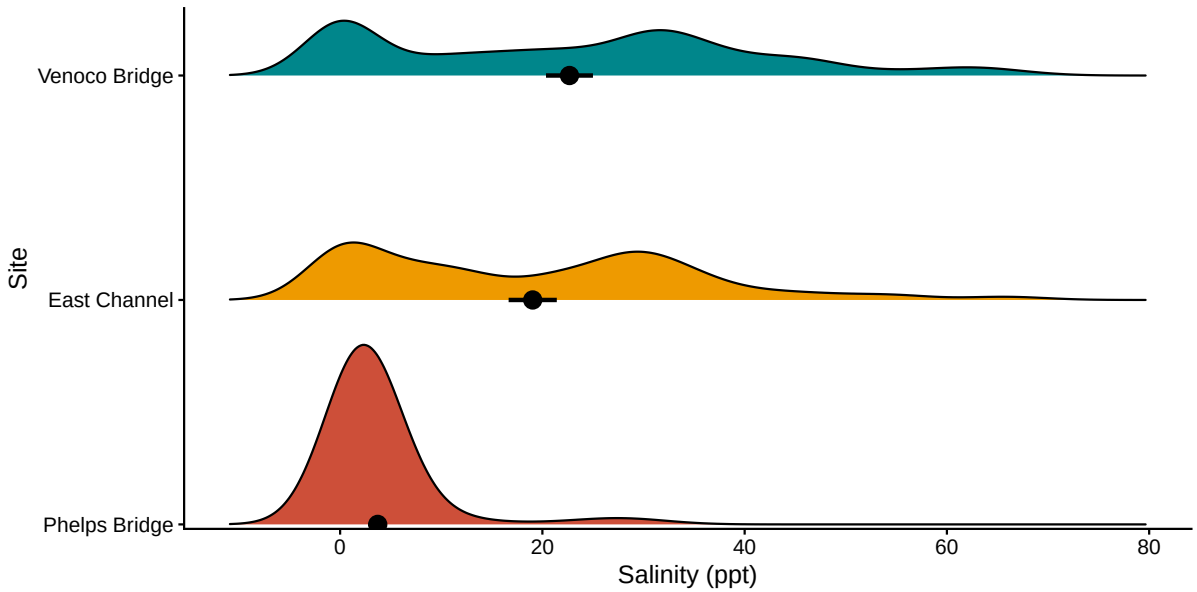
Figure 10. Mean salinity differs across sites. Small open circles represent observations of salinity at each sampling point at each site (red: Phelps Bridge, orange: East Channel, and teal: Venoco Bridge). Large points represent means. Whiskers represent 95% confidence intervals around the mean.

5. Making new types of plots and combining geometries

There are lots of options to visualize variables to compare across groups. Uuse graph galleries like the R Graph Gallery or From Data to Viz to come up with new ideas to try.

You can also combine different geometries and plots to represent data in a new way.

This plot is called a **ridgeline plot** because the distributions resemble mountain ridges. Adding points to represent means and 95% confidence intervals adds a metric of central tendency to the plot.



6. Saving figures

Sometimes it is useful to save figures as files, either to test out the dimensions or to use the figures elsewhere.

You can save a figure as any file type (e.g. .jpg, .png, .svg, .pdf) using `ggsave()`.

7. Analyses to compare metrics of central tendency across groups

Medians

if comparing two groups: **Mann Whitney U**

if comparing more than two groups: **Kruskal-Wallis**

Means

if comparing two groups: **two-sample t-test**

if comparing more than two groups: **analysis of variance (ANOVA)**

8. Project work

Get with your group members and discuss:

- which of the plots above would be appropriate for your variables
- which - if any - of these plots map onto the potential figures you designed in your work plan (if they don't, modify one of your potential figures to include a comparison of groups)
- (if your main question is comparing a response variable across groups) decide which metric of central tendency and test you should be using (from part 6 of this sheet)

Then,

- (if you haven't already) create a repo for your final project and make sure everyone is added as a collaborator
- create `code` and `data` folders
- create a file in your `code` folder for *drafts* of your data visualization
- work together to try 1-2 of the above plots
- work together to try 1-2 different plots from the R Graph Gallery or From Data to Viz

> ℹ All final projects must include at least one figure in which you are comparing your response variable across groups.
>
> A figure in which you are comparing your response variable across groups might directly answer your question or might provide context for your question - either way, you should have at least *one* figure comparing a metric of central tendency across groups.